

Technical Data Sheet: KNSB Propellant Grain Design Profile

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Abstract

This specification document establishes the nominal design layout and theoretical performance metrics for a standard Potassium Nitrate and Sorbitol (KNSB) solid propellant grain. This configuration serves as the perfect mathematical baseline, assuming a 100% efficient thermal dehydration cycle that removes all bound and free water content from the initial liquid binder charge. This profile is used by the R&D team as an analytical benchmark to evaluate the efficiency of physical batches processed in the lab.

Section 1: Ideal Batch Specifications Checklist

The theoretical mass breakdowns, formulation ratios, and intended geometric constraints for the reference model are detailed in Table I.

Table I: Theoretical Batch Specifications and Metrics

Parameter / Identifier	Target Specification	Reference Metric
Experimental Role	Analytical Reference Design	Nominal Target Standard
Propellant Formulation	KNSB (65:35)	65.00% Oxidizer : 35.00% Fuel
Initial Fuel Mass (Sorbitol 70)	800.0 g	800.0 g (Input Baseline)
Theoretical Anhydrous Mass	560.0 g	560.0 g (0% Residual Moisture)

Required Oxidizer Mass (KNO₃)	1040.0 g	1040.0 g (Stoichiometric Mass)
Slurry Blending Temperature	90°C	90°C (Thermal Equilibrium)
Target Casting Temperature	60°C	60°C (Optimal Fluidity)
Liner Tare Weight	Standard Baseline	145.0 g
Calculated Net Propellant Mass	1100.0 g	1100.0 g (Anhydrous Yield)
Total Assembly Gross Mass	1245.0 g	1245.0 g (Propellant + Liner)
Grain Outer Radius (\$R_o\$)	8.60 cm	8.60 cm
Grain Internal Core Radius (\$R_i\$)	3.55 cm	3.55 cm
Grain Column Length (\$L\$)	14.50 cm	14.50 cm
Density	1.84g/cm ³	1.566 g/cm ³

